Reinforcement Learning

Naeemullah Khan

naeemullah.khan@kaust.edu.sa

KAUST Academy

King Abdullah University of Science and Technology

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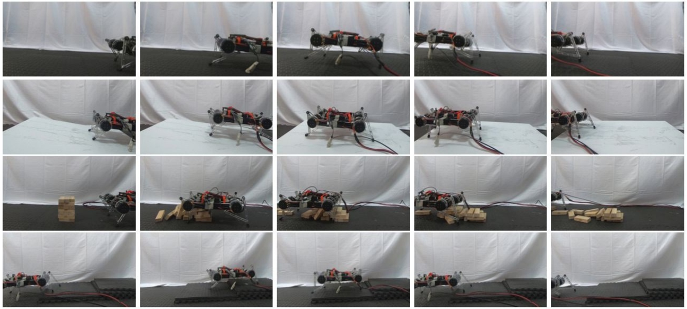
Soft Actor-Critic:
Off-Policy Maximum Entropy Deep Reinforcement



Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, Sergey Levine

Main Problem: Sample Inefficiency

- Number of times the agent must interact with the environment in order to learn a task
- Good sample complexity is the first prerequisite for successful skill acquisition.
- Learning skills in the real world can take a substantial amount of time
  - can get damaged through trial and error

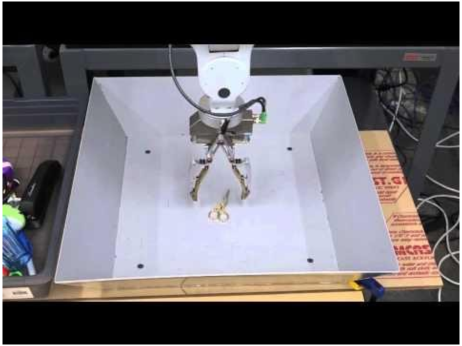
Main Problem: Sample Inefficiency

- "Learning Hand-Eye Coordination for Robotic Grasping with Deep Learning and Large-Scale Data Collection", Levine et al., 2016
  - 14 robot arms learning to grasp in parallel
  - objects started being picked up at around 20,000 grasps

https://spectrum.ieee.org/automaton/robotics/artificial-intelligence/google-large-scale-robotic-grasping-project

Main Problem: Sample Inefficiency

Observing the behavior of the robot after over 800,000 grasp attempts, which is equivalent to about 3000 robot-hours of practice, we can see the beginnings of intelligent reactive behaviors.



https://www.youtube.com/watch?v=cXaic_k80uM

Main Problem: Sample Inefficiency

- Solution?
- Off-Policy Learning!

Background: On-Policy vs. Off-Policy

- **On-policy learning:** use the deterministic outcomes or samples from the target policy to train the algorithm
  - has low sample efficiency (TRPO, PPO, A3C)
  - require new samples to be collected for nearly every update to the policy
  - becomes extremely expensive when the task is complex
- **Off-policy methods:** training on a distribution of transitions or episodes produced by a different behavior policy rather than that produced by the target policy
  - Does not require full trajectories and can reuse any past episodes (experience replay) for much better sample efficiency
  - relatively straightforward for Q-learning based methods

Background: Bellman Equation

- **Value Function:** How good is a state?

$$
\begin{aligned}
V(s) &= \mathbb{E}[G_t \mid S_t = s] \\
&= \mathbb{E}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots \mid S_t = s] \\
&= \mathbb{E}[R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \ldots) \mid S_t = s] \\
&= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\
&= \mathbb{E}[\underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{temporal difference target}} \mid S_t = s]
\end{aligned}
$$

- Similarly, for **Q-Function:** How good is a state-action pair?

$$
\begin{aligned}
Q(s,a) &= \mathbb{E}[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s, A_t = a] \\
&= \mathbb{E}[R_{t+1} + \gamma\, \mathbb{E}_{a\sim\pi} Q(S_{t+1}, a) \mid S_t = s, A_t = a]
\end{aligned}
$$

Background: Value-Based Method

- $\ldots, S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, \ldots$ **(on-policy):**

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big(R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\big)$$

- **Q-Learning (off-policy)**

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big(R_{t+1} + \gamma \max_{a \in \mathcal{A}} Q(S_{t+1}, a) - Q(S_t, A_t)\big)$$

- **DQN**, Minh et al., 2015

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim U(D)}\left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta)\right)^2\right]$$

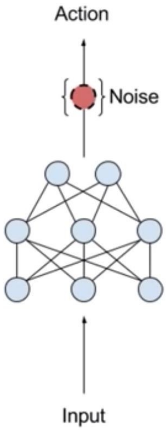
- Function Approximation
- Experience Replay: samples randomly drawn from replay memory
- Doesn't scale to continuous action space

Background: Policy-Based Method (Actor-Critic)

- **Critic:** updates value function parameters w and depending on the algorithm it could be action-value $Q(a \mid s; w)$ or state-value $V(s; w)$.
- **Actor:** updates policy parameters θ, in the direction suggested by the critic, $\pi(a \mid s; \theta)$.

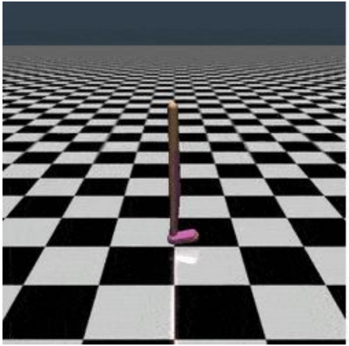
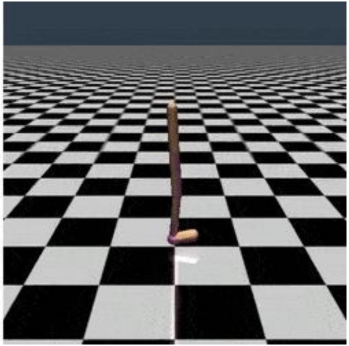
$$
\begin{aligned}
\nabla \mathcal{J}(\theta) &= \mathbb{E}_{\pi_\theta}[\nabla \ln \pi(a \mid s, \theta)\, Q_\pi(s, a)] && \text{policy gradient} \\
\theta &\leftarrow \theta + \alpha_\theta\, Q(s, a; w)\, \nabla_\theta \ln \pi(a \mid s; \theta) && \text{update actor} \\
G_{t:t+1} &= r_t + \gamma Q(s', a'; w) - Q(s, a; w) && \text{correction for action-value} \\
w &\leftarrow w + \alpha_w\, G_{t:t+1}\, \nabla_w Q(s, a; w) && \text{update critic}
\end{aligned}
$$

- DDPG = DQN + DPG (Lillicrap et al., 2015)
  - off-policy actor-critic method that learns a deterministic policy in continuous domain
  - exploration noise added to the deterministic policy when select action
  - difficult to stabilize and brittle to hyperparameters (Duan et al., 2016, Henderson et al., 2017)
  - unscalable to complex tasks with high dimensions (Gu et al., 2017)



https://www.youtube.com/watch?v=zR11FLZ-O9M

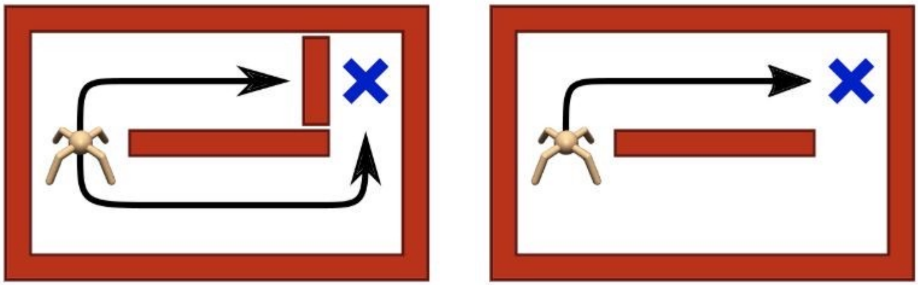
- Training is sensitive to randomness in the environment, initialization of the policy and the algorithm implementation

https://gym.openai.com/envs/Walker2d-v2/

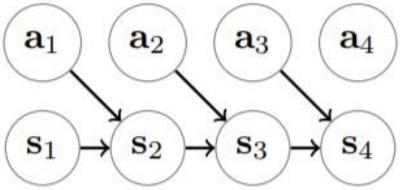
Main Problems: Robustness

- Knowing only one way to act makes agents vulnerable to environmental changes that are common in the real-world

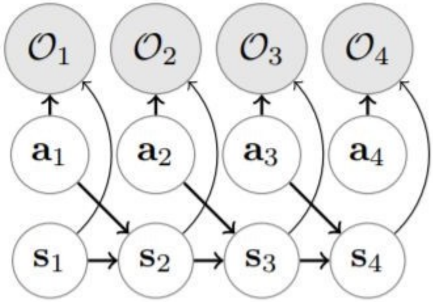


https://bair.berkeley.edu/blog/2017/10/06/soft-q-learning/

Background: Control as Inference



Traditional Graph of MDP



Graphical Model with Optimality Variables

Normal trajectory distribution

$$p(\tau) = p(\mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_T, \mathbf{a}_T \mid \theta) = p(\mathbf{s}_1) \prod_{t=1}^{T} p(\mathbf{a}_t \mid \mathbf{s}_t, \theta)\, p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t).$$

Posterior trajectory distribution

$$p(\mathcal{O}_t = 1 \mid \mathbf{s}_t, \mathbf{a}_t) = \exp(r(\mathbf{s}_t, \mathbf{a}_t)).$$

$$\begin{aligned} p(\tau \mid \mathbf{o}_{1:T}) \propto p(\tau, \mathbf{o}_{1:T}) &= p(\mathbf{s}_1) \prod_{t=1}^{T} p(\mathcal{O}_t = 1 \mid \mathbf{s}_t, \mathbf{a}_t)\, p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) \\ &= p(\mathbf{s}_1) \prod_{t=1}^{T} \exp(r(\mathbf{s}_t, \mathbf{a}_t))\, p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) \\ &= \left[p(\mathbf{s}_1) \prod_{t=1}^{T} p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) \right] \exp\left(\sum_{t=1}^{T} r(\mathbf{s}_t, \mathbf{a}_t) \right). \end{aligned}$$

Variational Inference

$$D_{\mathrm{KL}}(\hat{p}(\tau)\|p(\tau)) = -\mathbb{E}_{\tau\sim\hat{p}(\tau)}[\log p(\tau) - \log \hat{p}(\tau)].$$

$$\begin{aligned}
-D_{\mathrm{KL}}(\hat{p}(\tau)\|p(\tau)) &= \mathbb{E}_{\tau\sim\hat{p}(\tau)}\Bigg[\log p(\mathbf{s}_1) + \sum_{t=1}^{T}\Big(\log p(\mathbf{s}_{t+1}\mid\mathbf{s}_t,\mathbf{a}_t) + r(\mathbf{s}_t,\mathbf{a}_t)\Big) \\
&\qquad - \log p(\mathbf{s}_1) - \sum_{t=1}^{T}\Big(\log p(\mathbf{s}_{t+1}\mid\mathbf{s}_t,\mathbf{a}_t) + \log\pi(\mathbf{a}_t\mid\mathbf{s}_t)\Big)\Bigg] \\
&= \mathbb{E}_{\tau\sim\hat{p}(\tau)}\left[\sum_{t=1}^{T} r(\mathbf{s}_t,\mathbf{a}_t) - \log\pi(\mathbf{a}_t\mid\mathbf{s}_t)\right] \\
&= \sum_{t=1}^{T}\mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\hat{p}(\mathbf{s}_t,\mathbf{a}_t)}[r(\mathbf{s}_t,\mathbf{a}_t) - \log\pi(\mathbf{a}_t\mid\mathbf{s}_t)] \\
&= \sum_{t=1}^{T}\mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\hat{p}(\mathbf{s}_t,\mathbf{a}_t)}[r(\mathbf{s}_t,\mathbf{a}_t)] + \mathbb{E}_{\mathbf{s}_t\sim\hat{p}(\mathbf{s}_t)}[\mathcal{H}(\pi(\mathbf{a}_t\mid\mathbf{s}_t))].
\end{aligned}$$

Conventional RL Objective — Expected Reward

$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_t,\mathbf{a}_t)]$$

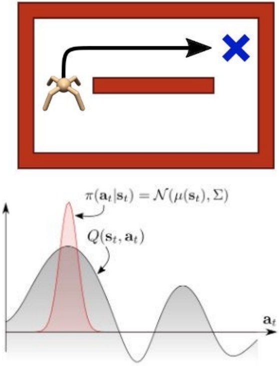
Maximum Entropy RL Objective — Expected Reward + Entropy of Policy

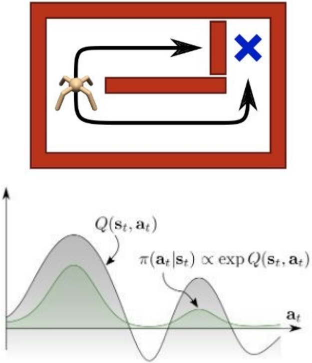
$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_t,\mathbf{a}_t) + \alpha\,\mathcal{H}(\pi(\cdot \mid \mathbf{s}_t))]$$

Entropy of a RV x

$$H(P) = \mathbb{E}_{x\sim P}[-\log P(x)]$$

- MaxEnt RL agent can capture different modes of optimality to improve robustness against environmental changes





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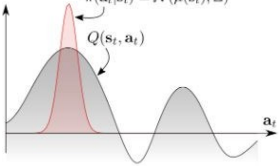
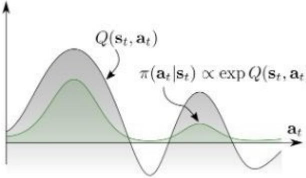




$$
\begin{aligned}
&\min_{\pi} D_{\mathrm{KL}}\left(\pi(\cdot \mid s_0) \,\|\, \exp(Q(s_0, \cdot))\right) \\
&= \min_{\pi} \mathbb{E}_{\pi}\left[\log \frac{\pi(a_0 \mid s_0)}{\exp(Q(s_0, a_0))}\right] \\
&= \max_{\pi} \mathbb{E}_{\pi}[Q(s_0, a_0) - \log \pi(a_0 \mid s_0)] \\
&= \max_{\pi} \mathbb{E}_{\pi}\left[\sum_t r(s_t, a_t) + \mathcal{H}(\pi(\cdot \mid s_0)) \mid s_0\right] \\
&= J_{\mathrm{MaxEnt}}(\pi(\cdot \mid s_0))
\end{aligned}
$$

Prior Work: Soft Q-Learning

- **Soft Q-Learning** (Haarnoja et al., 2017)
  - off-policy algorithms under MaxEnt RL objective
  - Learns Q^* directly
  - Instead of taking max action, sample the action
  - use approximate inference methods to sample
    - Stein variational gradient descent
  - not true actor-critic

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \Big(R_{t+1} + \gamma \max_{a \in \mathcal{A}} Q(S_{t+1}, a) - Q(S_t, A_t) \Big)$$

$$Q_{\text{soft}}(\mathbf{s}_t, \mathbf{a}_t) \leftarrow r_t + \gamma\, \mathbb{E}_{\mathbf{s}_{t+1} \sim p_{\mathbf{s}}} \left[V_{\text{soft}}(\mathbf{s}_{t+1}) \right], \quad \forall \mathbf{s}_t, \mathbf{a}_t$$

$$V_{\text{soft}}(\mathbf{s}_t) \leftarrow \alpha \log \int_{\mathcal{A}} \exp\left(\tfrac{1}{\alpha} Q_{\text{soft}}(\mathbf{s}_t, \mathbf{a}') \right) d\mathbf{a}', \quad \forall \mathbf{s}_t$$

SAC: Contributions

- One of the most efficient model-free algorithms
  - SOTA off-policy
  - well suited for real world robotics learning
- Can learn stochastic policy on continuous action domain
- Robust to noise
- **Ingredients:**
  - Actor-critic architecture with seperate policy and value function networks
  - Off-policy formulation to reuse of previously collected data for efficiency
  - Entropy-constrained objective to encourage stability and exploration

- **policy evaluation:** compute value of π according to Max Entropy RL Objective
- modified Bellman backup operator $\mathcal{T}$:

$$\mathcal{T}^{\pi} Q(\mathbf{s}_t, \mathbf{a}_t) \triangleq r(\mathbf{s}_t, \mathbf{a}_t) + \gamma\, \mathbb{E}_{\mathbf{s}_{t+1} \sim p}\left[\boxed{V(\mathbf{s}_{t+1})}\right]$$

$$\boxed{V(\mathbf{s}_t)} = \mathbb{E}_{\mathbf{a}_t \sim \pi}\left[Q(\mathbf{s}_t, \mathbf{a}_t) - \boxed{\alpha \log \pi(\mathbf{a}_t \mid \mathbf{s}_t)}\right]$$

- **Lemma 1:** Contraction Mapping for Soft Bellman Updates

$$Q^{k+1} = \mathcal{T}^{\pi} Q^{k} \quad \text{converges to the soft Q-function of } \pi$$

Soft Policy Iteration: Policy Improvement

- **policy improvement:** update policy towards the exponential of the new soft Q-function
- modified Bellman backup operator $\mathcal{T}$:
  - choose tractable family of distributions big Π
  - choose KL divergence to project the improved policy into big Π

$$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} D_{\mathrm{KL}}\left(\pi'(\,\cdot \mid \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(\frac{1}{\alpha} Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right)}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)}\right)$$

- **Lemma 2**

$$Q^{\pi_{\text{new}}}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi_{\text{old}}}(\mathbf{s}_t, \mathbf{a}_t) \quad \text{for any state action pair}$$

- soft policy iteration: soft policy evaluation $\longleftrightarrow$ soft policy improvement
- **Theorem 1:** Repeated application of soft policy evaluation and soft policy improvement from any policy $\pi \in \Pi$ converges to the optimal MaxEnt policy among all policies in Π
    - exact form applicable only in discrete case
    - need function approximation to represent Q-values in continuous domains
    - $\rightarrow$ Soft Actor-Critic (SAC)!

SAC — Parameterized soft Q-function

$Q_\theta(\mathbf{s}_t, \mathbf{a}_t)$

- e.g. neural network

$\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)$

parameterized tractable policy

- e.g. Gaussian with mean and covariances given by neural networks

$J_Q(\theta) \quad \hat{\nabla}_\theta J_Q(\theta)$

soft Q-function objective and its stochastic gradient wrt its parameters

$J_\pi(\phi) \quad \hat{\nabla}_\phi J_\pi(\phi)$

policy objective and stochastic gradient wrt its parameters

- **Critic — Soft Q-function**
  - minimize square error
  - $\bar{\theta}$ exponential moving average of soft Q-function weights to stabilize training (DQN)

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}} \left[\frac{1}{2} \Big(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \big(r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p}[V_{\bar{\theta}}(\mathbf{s}_{t+1})] \big) \Big)^2 \right]$$

$$V(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t \sim \pi}[Q(\mathbf{s}_t, \mathbf{a}_t) - \alpha \log \pi(\mathbf{a}_t \mid \mathbf{s}_t)]$$

$$\hat{\nabla}_\theta J_Q(\theta) = \nabla_\theta Q_\theta(\mathbf{a}_t, \mathbf{s}_t) \big(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \big(r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \big(Q_{\bar{\theta}}(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) - \alpha \log(\pi_\phi(\mathbf{a}_{t+1} \mid \mathbf{s}_{t+1})) \big) \big) \big)$$

- **Actor — Policy** $$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} D_{\mathrm{KL}}\left(\pi'(\,\cdot \mid \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(\frac{1}{\alpha} Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right)}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)}\right)$$
- multiply by alpha and ignoring the normalization Z

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}}\left[\mathbb{E}_{\mathbf{a}_t \sim \pi_\phi}[\alpha \log(\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t)]\right]$$

- reparameterize with neural network f $\quad \mathbf{a}_t = f_\phi(\epsilon_t; \mathbf{s}_t)$
  - epsilon: input noise vector, sampled from a fixed distribution (spherical Gaussian)

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D},\, \epsilon_t \sim \mathcal{N}}[\alpha \log \pi_\phi(f_\phi(\epsilon_t; \mathbf{s}_t) \mid \mathbf{s}_t) - Q_\theta(\mathbf{s}_t, f_\phi(\epsilon_t; \mathbf{s}_t))]$$

$$\hat{\nabla}_\phi J_\pi(\phi) = \nabla_\phi \alpha \log(\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)) + (\nabla_{\mathbf{a}_t} \alpha \log(\pi_\phi(\mathbf{a}_t \mid \mathbf{s}_t)) - \nabla_{\mathbf{a}_t} Q(\mathbf{s}_t, \mathbf{a}_t)) \nabla_\phi f_\phi(\epsilon_t; \mathbf{s}_t)$$

- **Unbiased gradient estimator that extends DDPG stype policy gradients to any tractable stochastic policy**

SAC: Algorithm

Algorithm 1 Soft Actor-Critic

Input: initial policy parameters θ, Q-function parameters ϕ_1, ϕ_2, empty replay buffer $\mathcal{D}$
Set target parameters equal to main parameters $\phi_{\text{targ},1} \leftarrow \phi_1$, $\phi_{\text{targ},2} \leftarrow \phi_2$
repeat
  Observe state s and select action $a \sim \pi_\theta(\cdot|s)$
  Execute a in the environment
  Observe next state s', reward r, and done signal d to indicate whether s' is terminal
  Store (s, a, r, s', d) in replay buffer $\mathcal{D}$
  If s' is terminal, reset environment state.
  if it's time to update **then**
    for j in range(however many updates) **do**
      Randomly sample a batch of transitions, $B = \{(s, a, r, s', d)\}$ from $\mathcal{D}$
      Compute targets for the Q functions:

$$y(r, s', d) = r + \gamma(1-d)\left(\min_{i=1,2} Q_{\phi_{\text{targ},i}}(s', \tilde{a}') - \alpha \log \pi_\theta(\tilde{a}'|s')\right), \quad \tilde{a}' \sim \pi_\theta(\cdot|s')$$

      Update Q-functions by one step of gradient descent using

$$\nabla_{\phi_i} \frac{1}{|B|} \sum_{(s,a,r,s',d)\in B} \left(Q_{\phi_i}(s,a) - y(r,s',d)\right)^2 \qquad \text{for } i = 1, 2$$

      Update policy by one step of gradient ascent using

$$\nabla_\theta \frac{1}{|B|} \sum_{s\in B} \left(\min_{i=1,2} Q_{\phi_i}(s, \tilde{a}_\theta(s)) - \alpha \log \pi_\theta\left(\tilde{a}_\theta(s)|\, s\right)\right),$$

      where $\tilde{a}_\theta(s)$ is a sample from $\pi_\theta(\cdot|s)$ which is differentiable wrt θ via the reparametrization trick.
      Update target networks with

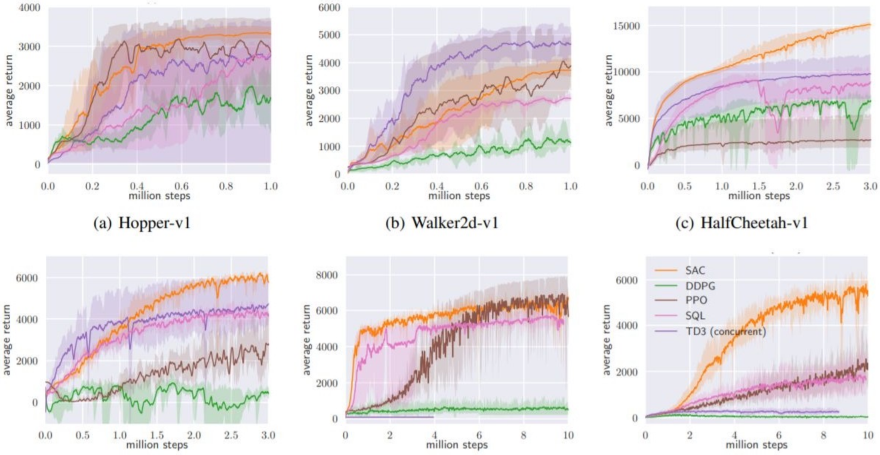
$$\phi_{\text{targ},i} \leftarrow \rho\phi_{\text{targ},i} + (1-\rho)\phi_i \qquad \text{for } i = 1, 2$$

    end for
  end if
until convergence

https://spinningup.openai.com/en/latest/algorithms/sac.html

- **Tasks**
  - A range of continuous control tasks from the OpenAI gym benchmark suite
  - RL-Lab implementation of the Humanoid task
  - The easier tasks can be solved by a wide range of different algorithms, the more complex benchmarks, such as the 21-dimensional Humanoid (rllab) are exceptionally difficult to solve with off-policy algorithms.
- **Baselines:**
  - DDPG, SQL, PPO, TD3 (concurrent)
  - TD3 is an extension to DDPG that first applied the double Q-learning trick to continuous control along with other improvements.

SAC: Results

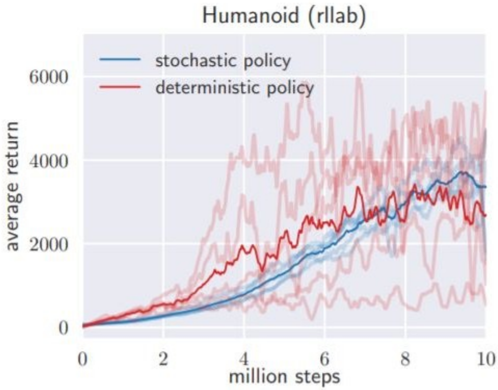


(a) Hopper-v1 (b) Walker2d-v1 (c) HalfCheetah-v1

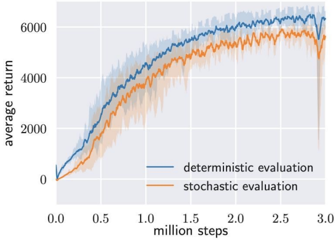
(d) Ant-v1 (e) Humanoid-v1 (f) Humanoid (rllab)

Experimental Results: Ablation Study

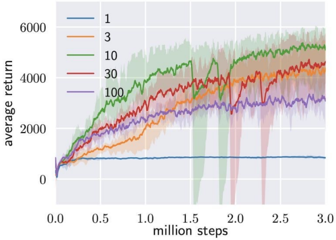
- How does the stochasticity of the policy and entropy maximization affect the performance?
- Comparison with a deterministic variant of SAC that does not maximize the entropy and that closely resembles DDPG



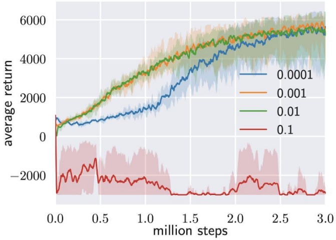
https://arxiv.org/abs/1801.01290



(a) Evaluation



(b) Reward Scale



(c) Target Smoothing Coefficient (τ)

https://arxiv.org/abs/1801.01290

Limitation

- Unfortunately, SAC also suffers from brittleness to the alpha temperature hyperparameter that controls exploration
  - $\rightarrow$ automatic temperature tuning!

Contributions

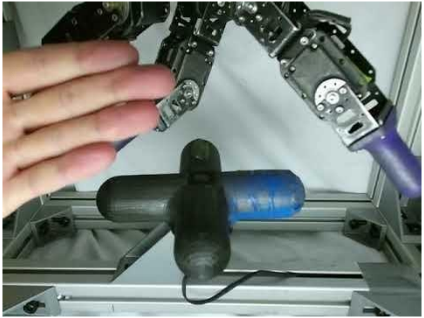
- Adaptive temperature coefficient

$$\sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}\left[r(\mathbf{s}_t,\mathbf{a}_t)+\underbrace{\alpha}_{\text{temperature}}\mathcal{H}(\pi(\cdot\mid\mathbf{s}_t))\right]$$

- Extend to real-world tasks such as locomotion for a quadrupedal robot and robotic manipulation with a dexterous hand

https://arxiv.org/abs/1801.01290

- Dexterous Hand Manipulations
- 20 hour end-to-end learning
- valve position as input: SAC 3 hours vs. PPO 7.4 hours



https://sites.google.com/view/sac-and-applications

- Choosing the optimal temperature is non-trivial (tuned for each task)
- Constrained optimization problem:

$$\max \sum_t \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_t,\mathbf{a}_t) + \alpha\mathcal{H}(\pi(\cdot \mid \mathbf{s}_t))]$$

$$\max_{\pi_{0:T}} \mathbb{E}_{\rho_\pi}\left[\sum_{t=0}^{T} r(\mathbf{s}_t,\mathbf{a}_t)\right] \quad \text{s.t.} \quad \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[-\log(\pi_t(\mathbf{a}_t \mid \mathbf{s}_t))] \geq \mathcal{H} \quad \forall t$$

Unroll the expectation

$$\max_{\pi_0}\left(\mathbb{E}[r(\mathbf{s}_0,\mathbf{a}_0)]+\max_{\pi_1}\left(\mathbb{E}[\ldots]+\boxed{\max_{\pi_T}\mathbb{E}[r(\mathbf{s}_T,\mathbf{a}_T)]}\right)\right)$$

For the last time step in the trajectory

$$\max_{\pi_T}\mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\rho_\pi}[r(\mathbf{s}_T,\mathbf{a}_T)] = \min_{\alpha_T\geq 0}\max_{\pi_T}\mathbb{E}[r(\mathbf{s}_T,\mathbf{a}_T)-\alpha_T\log\pi(\mathbf{a}_T\mid\mathbf{s}_T)]-\alpha$$

$$\arg\min_{\alpha_T}\mathbb{E}_{\mathbf{s}_t,\mathbf{a}_t\sim\pi_t^*}[-\alpha_T\log\pi_T^*(\mathbf{a}_T\mid\mathbf{s}_T;\alpha_T)-\alpha_T\mathcal{H}]\,.$$

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Similarly, for the previous time step

$$\max_{\pi_{T-1}}\Big(\mathbb{E}[r(\mathbf{s}_{T-1},\mathbf{a}_{T-1})] + \max_{\pi_T}\mathbb{E}[r(\mathbf{s}_T,\mathbf{a}_T)]\Big) \tag{16}$$

$$= \max_{\pi_{T-1}}\big(Q^*_{T-1}(\mathbf{s}_{T-1},\mathbf{a}_{T-1}) - \alpha_T\mathcal{H}\big)$$

$$= \min_{\alpha_{T-1}\geq 0}\max_{\pi_{T-1}}\Big(\mathbb{E}[Q^*_{T-1}(\mathbf{s}_{T-1},\mathbf{a}_{T-1})] - \mathbb{E}[\alpha_{T-1}\log\pi(\mathbf{a}_{T-1}\mid\mathbf{s}_{T-1})] - \alpha_{T-1}\mathcal{H}\Big)$$

$$\alpha_t^* = \arg\min_{\alpha_t}\mathbb{E}_{\mathbf{a}_t\sim\pi_t^*}\left[-\alpha_t\log\pi_t^*(\mathbf{a}_t\mid\mathbf{s}_t;\alpha_t) - \alpha_t\bar{\mathcal{H}}\right]$$

Algorithm 1 Soft Actor-Critic

Input: θ_1, θ_2, ϕ two soft Q-functions ▷ Initial parameters

$\bar{\theta}_1 \leftarrow \theta_1, \bar{\theta}_2 \leftarrow \theta_2$ ▷ Initialize target network weights

$\mathcal{D} \leftarrow \emptyset$ ▷ Initialize an empty replay pool

for each iteration **do**

  for each environment step **do**

    $\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$ ▷ Sample action from the policy

    $\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$ ▷ Sample transition from the environment

    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$ ▷ Store the transition in the replay pool

  end for

  for each gradient step **do**

    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$ ▷ Update the Q-function parameters

    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$ ▷ Update policy weights

    $\alpha \leftarrow \alpha - \lambda \hat{\nabla}_\alpha J(\alpha)$ ▷ Adjust temperature

    $\bar{\theta}_i \leftarrow \tau\theta_i + (1 - \tau)\bar{\theta}_i$ for $i \in \{1, 2\}$ ▷ Update target network weights

  end for exponential moving average

end for

Output: θ_1, θ_2, ϕ ▷ Optimized parameters

Experimental Results: RL Lab

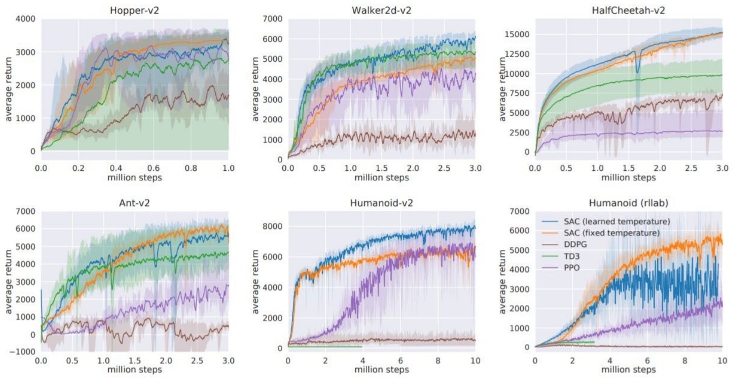

Figure 1: Training curves on continuous control benchmarks. Soft actor-critic (blue and yellow) performs consistently across all tasks and outperforming both on-policy and off-policy methods in the most challenging tasks.

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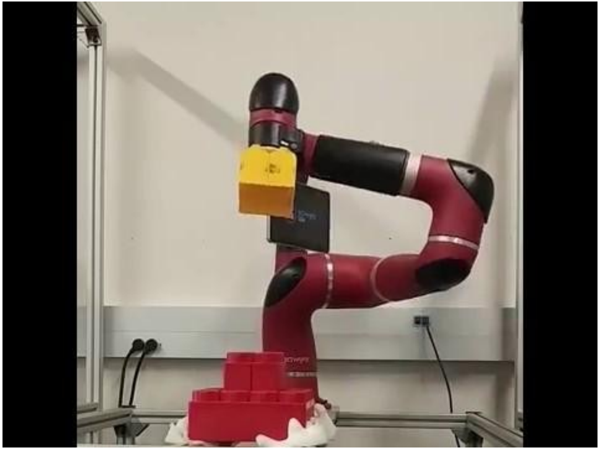
Experimental Results: Robustness



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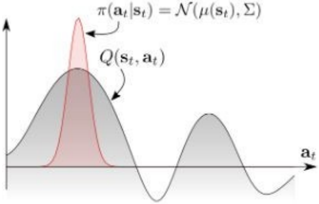

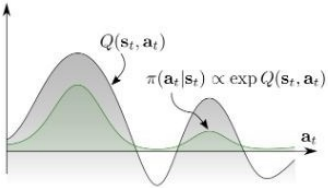

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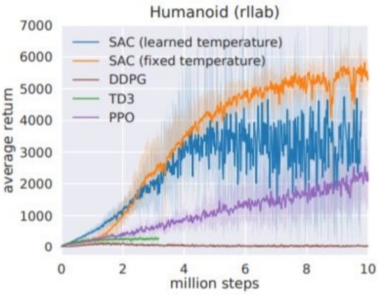

Recap: SAC

- An off-policy maximum entropy deep reinforcement learning algorithm
  - Sample-efficient
  - Scale to high-dimensional observation/action space
  - Robustness to random seed, noise and etc.
- Theoretical Results
  - Convergence of soft policy iteration
  - Derivation of soft-actor critic algorithm
- Empirical Results
  - SAC outperforms SOTA model-free deep RL methods, including DDPG, PPO and Soft Q-learning, in terms of the policy's optimality, sample complexity and robustness.

References

These slides have been adapted from

- Animesh Garg, CSC2621: Reinforcement Learning in Robotics, University of Toronto